C1 Problems: functions, graphs and equations

← Module README · Notes · Solutions

Work exactly unless a problem says you may use a graphing calculator (GR). “Bereken exact” means: no decimal approximations.

Practice

1. Give the domain and the range of

a. $f(x) = 3 - \sqrt{8 - 4x}$

b. $g(x) = \dfrac{1}{x^2 - 9}$

2. Given $g(x) = \dfrac{3x - 2}{x + 1}$.

a. Give the domain of g.
b. Give the equations of the asymptotes.
c. What is the range of g?

3. Describe a sequence of transformations that turns the graph of $y = \dfrac{1}{x}$ into the graph of $y = 2 - \dfrac{3}{x + 1}$.

4. The graph of $y = \sqrt{x}$ is first stretched from the y-axis by factor $\frac{1}{2}$ and then translated by $(3, -1)$.

a. Give the formula of the image.
b. Give the formula if the two transformations are done in the opposite order. Is the result the same graph?

5. Find the inverse function of $g(x) = \dfrac{3x - 2}{x + 1}$ and give its domain.

6. Solve exactly.

a. $|3x + 2| = 8$
b. $|3x + 2| \geq 8$

7. Solve exactly: $x^3 - x^2 - 6x > 0$.

8. Solve exactly, and check for false solutions.

a. $\sqrt{2x + 9} = x + 3$
b. $\sqrt{x + 5} = x - 1$

9. Given $f(x) = \dfrac{2x^2 + 1}{x^2 + 1}$.

a. Show that $f(x) = 2 - \dfrac{1}{x^2 + 1}$.
b. Give the horizontal asymptote and the range of f.

10. Decide whether each function is even, odd or neither. Prove your answer.

a. $f(x) = x^3 - x$ b. $g(x) = x^4 + x^2$ c. $h(x) = x^2 + x$ d. $k(x) = \dfrac{x}{x^2 + 1}$

11. Given $f(x) = x^2 + 6x + 5$ with domain $x \leq -3$. Find f^{-1} and its domain.

CE-level

12. Given $f(x) = \dfrac{x^2 - 9}{x^2 + x - 6}$.

a. Toon aan that the graph of f has exactly one vertical asymptote.
b. The graph of f has a hole (*perforatie*). Bereken exact its coordinates.
c. Give the horizontal asymptote.

13. Given $f(x) = \sqrt{x + 4}$ and $g(x) = \frac{1}{2}x + 2$.

a. Bereken exact the x-coordinates of the intersection points of the graphs of f and g.
b. Solve exactly: $f(x) > g(x)$.

14. Given $f(x) = \dfrac{1}{x}$ and the lines $\ell_p : y = -x + p$. Investigate, depending on p, how many intersection points the graph of f and ℓ_p have. Give the values of p for which ℓ_p touches the graph exactly.

15. Given $f(x) = 1 + \sqrt{x - 1}$.

a. Give the domain and range of f.
b. Toon aan that $f^{-1}(x) = (x - 1)^2 + 1$, and give the domain of f^{-1}.
c. Bereken exact the coordinates of the intersection points of the graphs of f and f^{-1}.

16. Given $f(x) = \dfrac{x^2 - 3x + 5}{x - 2}$.

a. Toon aan that $f(x) = x - 1 + \dfrac{3}{x - 2}$.
b. Give the equations of all asymptotes of the graph of f.

17. Solve exactly: $\dfrac{2x - 1}{x + 3} \leq 1$.

18. Solve exactly: $|x^2 - 2x| = x$.

19. Given $f(x) = x^2 + 4x$. The graph of g arises from the graph of f by a stretch from the y-axis by factor 3, followed by a translation $(0, -2)$.

a. Give the formula of g (expanded).
b. Find the vertex of the graph of g in two ways: from the formula, and by transforming the vertex of f.

20. The number of downloads (in thousands) of an app t weeks after launch is modelled by $N(t) = \dfrac{1200t}{t + 6}$, $t \geq 0$.

a. Which value does N approach in the long run? Explain.
b. Bereken exact from which moment $N > 900$.
c. Write t as a function of N.

Challenge

21. Given $f(x) = \dfrac{ax+b}{cx+d}$ with $c \neq 0$ and $ad - bc \neq 0$.

a. Find f^{-1}.
b. Prove that f is its own inverse ($f^{-1} = f$) if $d = -a$.
c. Why is the condition $ad - bc \neq 0$ needed?

22. Solve: $\sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{x-1}} = 1$. *Hint:* substitute $t = \sqrt{x-1}$.

23. Given $g(x) = |x^2 - 4|x| + 3|$. Find, for every real number k, the number of solutions of $g(x) = k$.

24. Let f be a strictly increasing function with inverse f^{-1}.

a. Prove: if the graphs of f and f^{-1} meet at (a, b), then $a = b$.
b. Give an example of a decreasing function f whose graph meets the graph of f^{-1} at a point not on the line $y = x$.

25. A function f defined for all $x \neq 0$ satisfies $f(x) + 2f\left(\frac{1}{x}\right) = 3x$ for all $x \neq 0$.

a. Find $f(x)$.
b. Solve $f(x) = f(-x)$.